

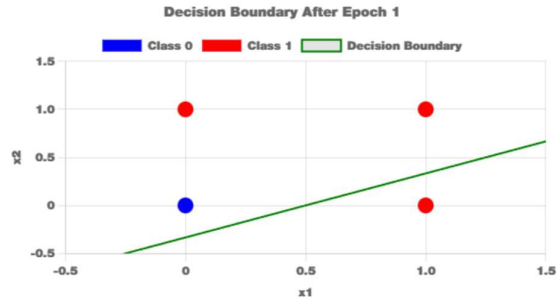
## OR Gate (Linearly Separable)

### OUTPUT:

#### ---EPOCH 1---

Accuracy after Epoch 1: 50.00%

Updated Weights ->  $w_1 = -0.200$ ,  $w_2 = 0.300$ ,  $b = 0.100$



#### ---EPOCH 2---

Accuracy after Epoch 2: 50.00%

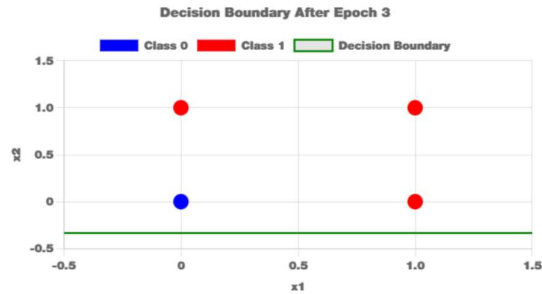
Updated Weights ->  $w_1 = -0.100$ ,  $w_2 = 0.300$ ,  $b = 0.100$



#### ---EPOCH 3---

Accuracy after Epoch 3: 50.00%

Updated Weights ->  $w_1 = 0.000$ ,  $w_2 = 0.300$ ,  $b = 0.100$



#### ---EPOCH 4---

Accuracy after Epoch 4: 50.00%

Updated Weights ->  $w_1 = 0.100$ ,  $w_2 = 0.300$ ,  $b = 0.100$



#### ---EPOCH 5---

Accuracy after Epoch 5: 75.00%

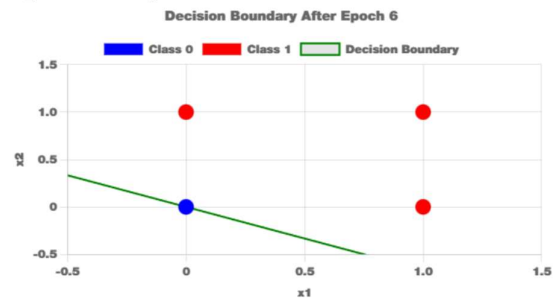
Updated Weights ->  $w_1 = 0.100$ ,  $w_2 = 0.300$ ,  $b = 0.000$



#### ---EPOCH 6---

Accuracy after Epoch 6: 50.00%

Updated Weights ->  $w_1 = 0.200$ ,  $w_2 = 0.300$ ,  $b = 0.000$



#### ---EPOCH 7---

Accuracy after Epoch 7: 75.00%

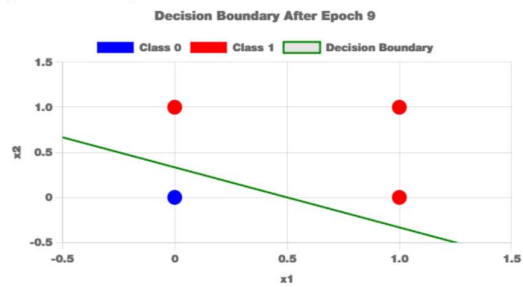
Updated Weights ->  $w_1=0.200$ ,  $w_2=0.300$ ,  $b=-0.100$



#### ---EPOCH 8---

Accuracy after Epoch 8: 100.00%

Updated Weights ->  $w_1=0.200$ ,  $w_2=0.300$ ,  $b=-0.100$



#### ---EPOCH 9---

Accuracy after Epoch 9: 100.00%

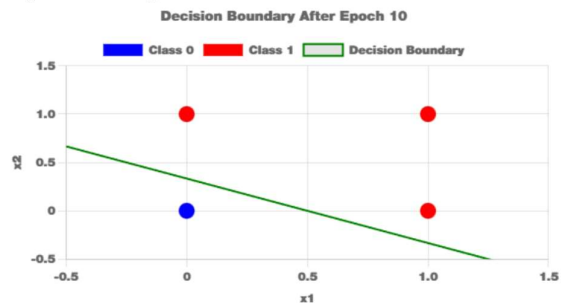
Updated Weights ->  $w_1=0.200$ ,  $w_2=0.300$ ,  $b=-0.100$



#### ---EPOCH 10---

Accuracy after Epoch 10: 100.00%

Updated Weights ->  $w_1=0.200$ ,  $w_2=0.300$ ,  $b=-0.100$



Final Accuracy:100.00%

| Input (x1, x2) | Predicted | Actual |
|----------------|-----------|--------|
| (0, 0)         | 0         | 0      |
| (0, 1)         | 1         | 1      |
| (1, 0)         | 1         | 1      |
| (1, 1)         | 1         | 1      |

### XOR Gate (Non-linearly Separable)

#### OUTPUT:

#### ---EPOCH 1---

Accuracy after Epoch 1: 50.00%

Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.801$ ,  $b=0.264$



#### ---EPOCH 2---

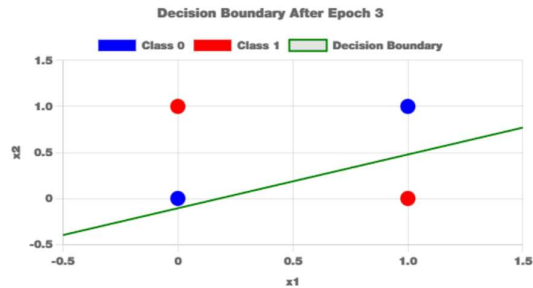
Accuracy after Epoch 2: 25.00%

Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.701$ ,  $b=0.164$



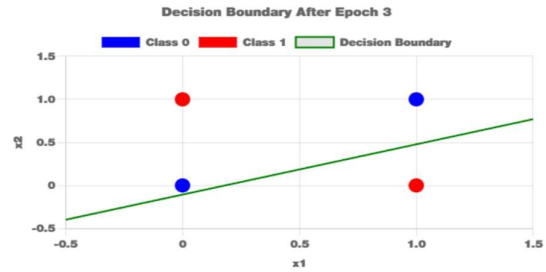
### ---EPOCH 3---

Accuracy after Epoch 3: 25.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.601$ ,  $b=0.064$



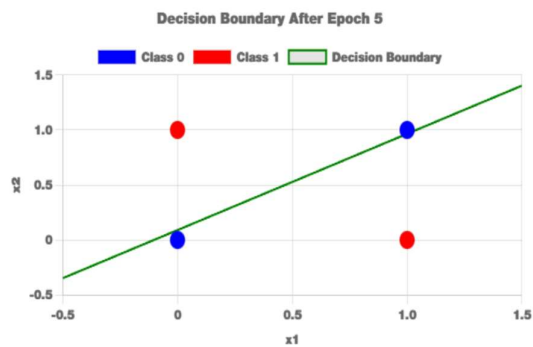
### ---EPOCH 4---

Accuracy after Epoch 4: 25.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.501$ ,  $b=-0.036$



### ---EPOCH 5---

Accuracy after Epoch 5: 50.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.401$ ,  $b=-0.036$



### ---EPOCH 6---

Accuracy after Epoch 6: 50.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.301$ ,  $b=-0.036$



### ---EPOCH 7---

Accuracy after Epoch 7: 50.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.201$ ,  $b=-0.036$



### ---EPOCH 8---

Accuracy after Epoch 8: 50.00%  
Updated Weights ->  $w_1=-0.351$ ,  $w_2=0.101$ ,  $b=-0.036$



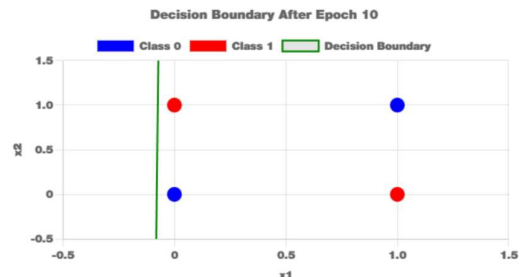
### ---EPOCH 9---

Accuracy after Epoch 9: 75.00%  
Updated Weights ->  $w_1=-0.251$ ,  $w_2=0.101$ ,  $b=0.064$



### ---EPOCH 10---

Accuracy after Epoch 10: 25.00%  
Updated Weights ->  $w_1=-0.251$ ,  $w_2=0.001$ ,  $b=-0.036$



Final Accuracy:50.00%

| Input (x1, x2) | Predicted | Actual |
|----------------|-----------|--------|
| (0,0)          | 0         | 0      |
| (0,1)          | 0         | 1      |
| (1,0)          | 0         | 1      |
| (1,1)          | 0         | 0      |